CAN

PROTOCOL

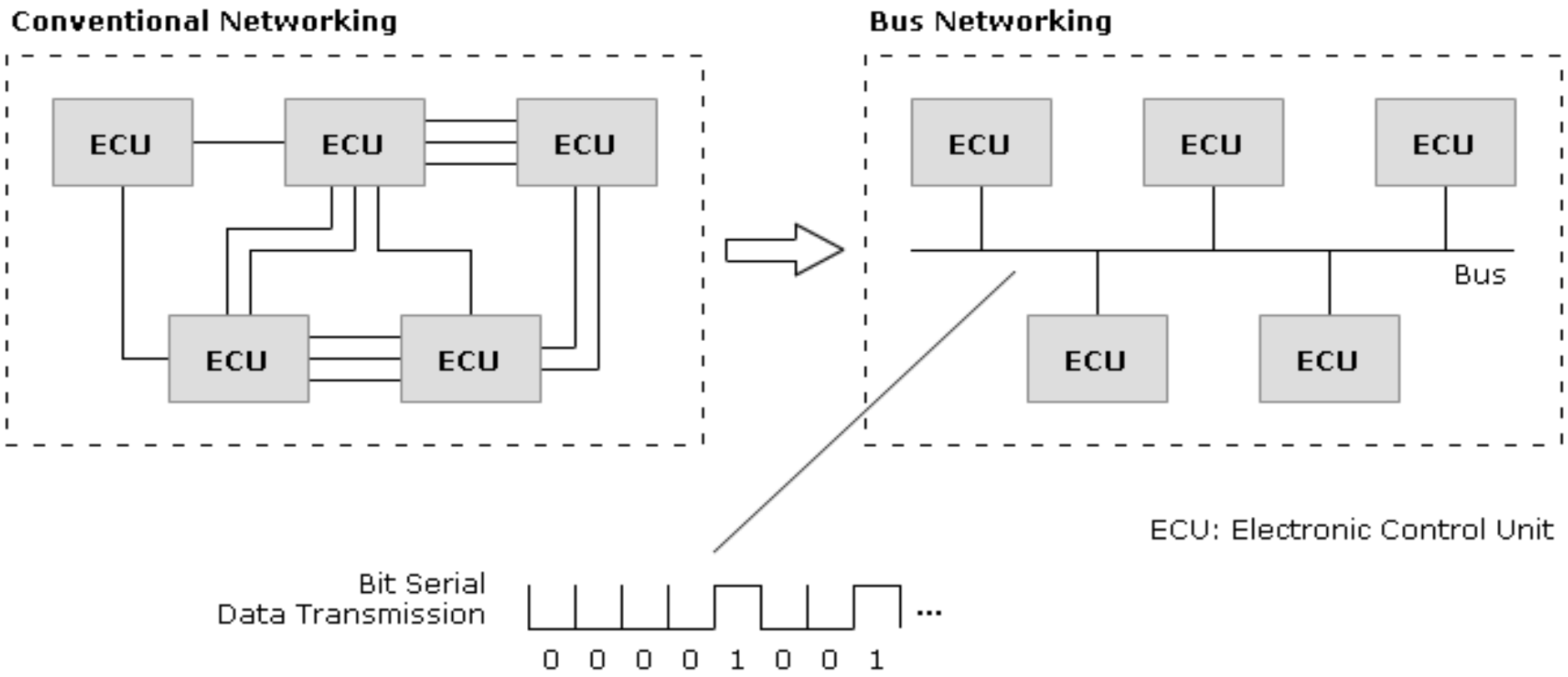
Features

1. The acronym “CAN” Stands for “Controller Area Network.
2. Developed by Robert Bosch,in1983,orginally for automobile in-vehicle network.
3. But has been adopted in other areas of engineering & technology also.

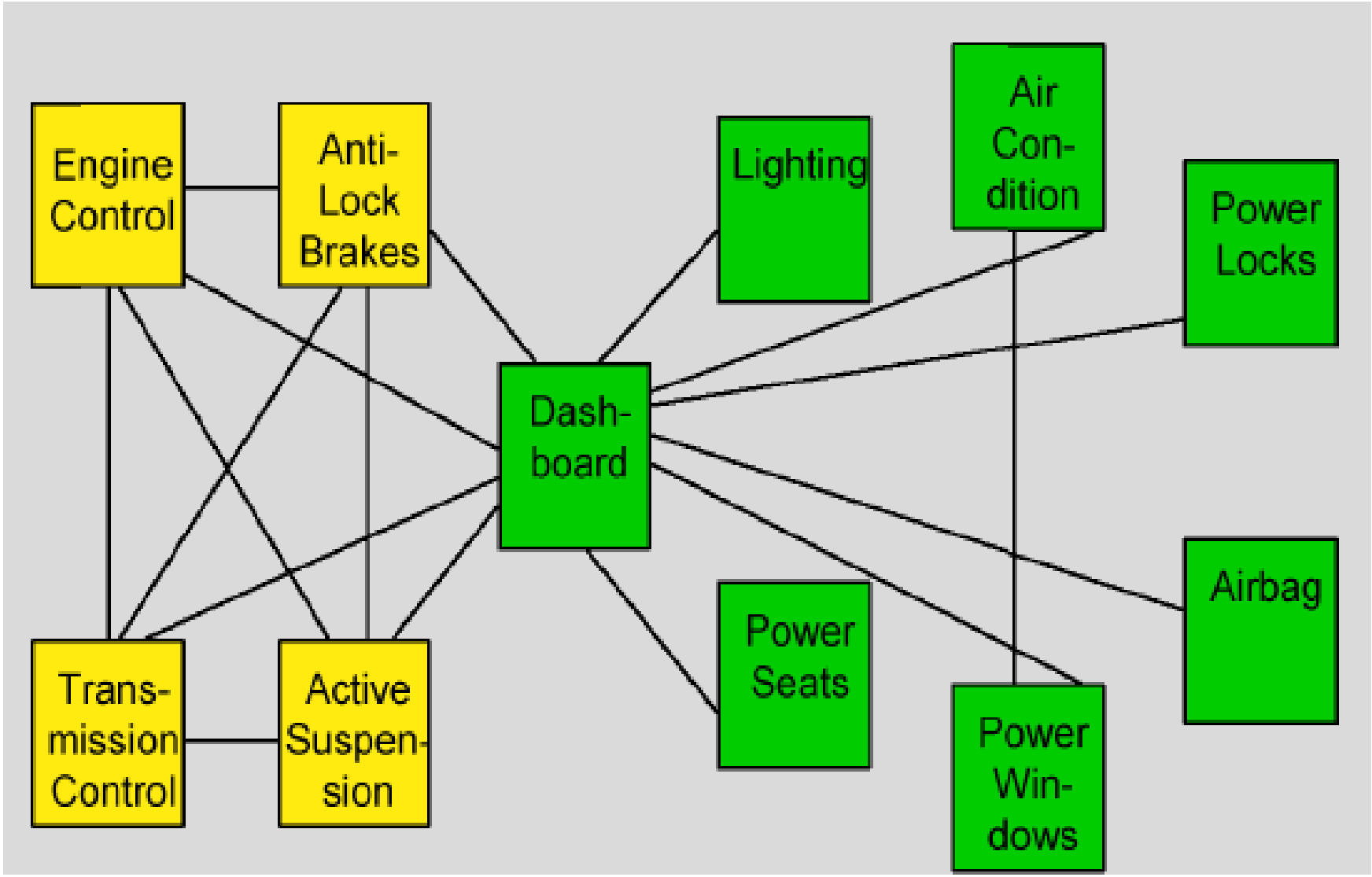


Features Cont'd

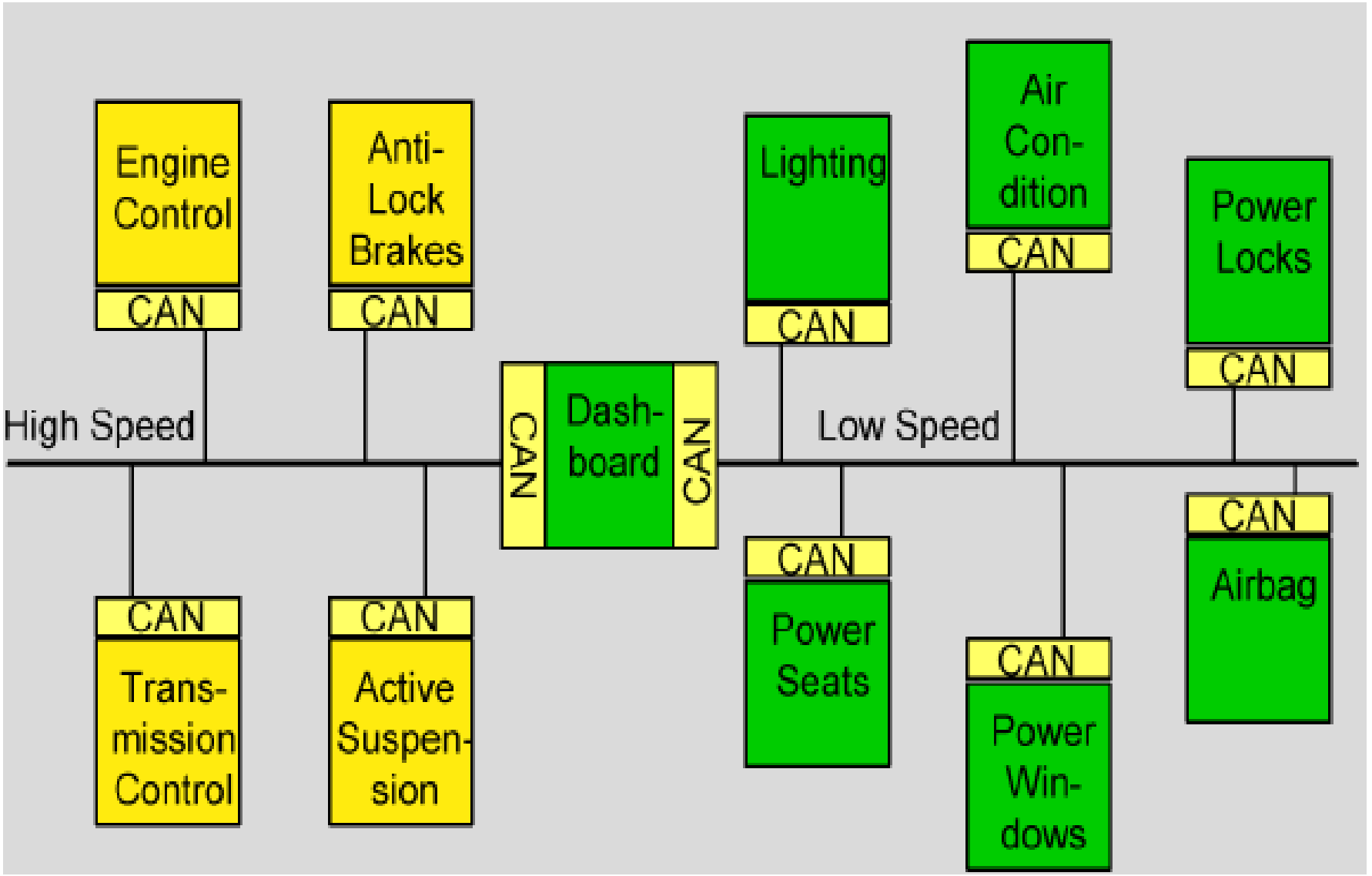
4. Supports multi-drop, multi-master-slave,serial bus communication between Node/ECUs in an automobile.



Before CAN

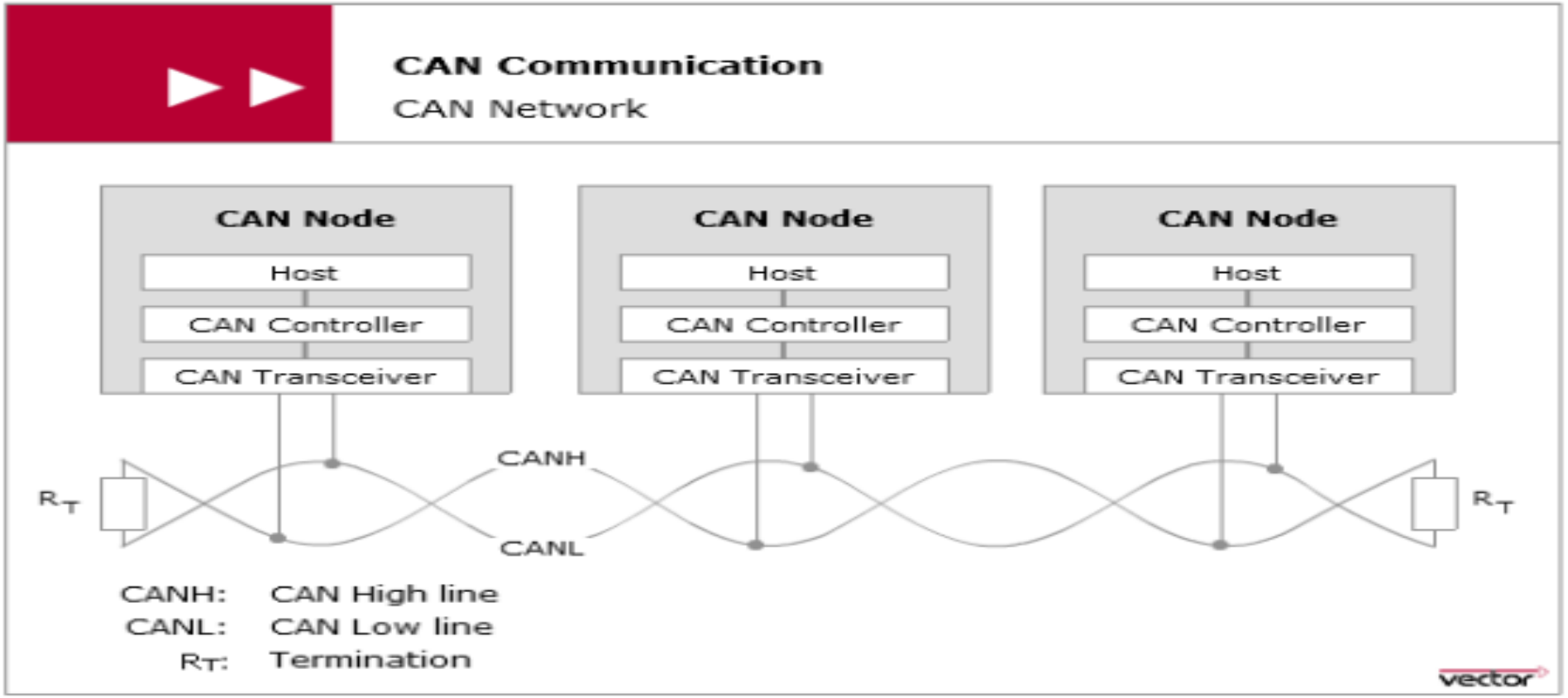

Engine Control
Anti-Lock Brakes
Lighting
Air Con-dition
Power Locks
Dash-board
Airbag
Power Seats
Trans-mission Control
Active Suspen-sion
Power Win-dows

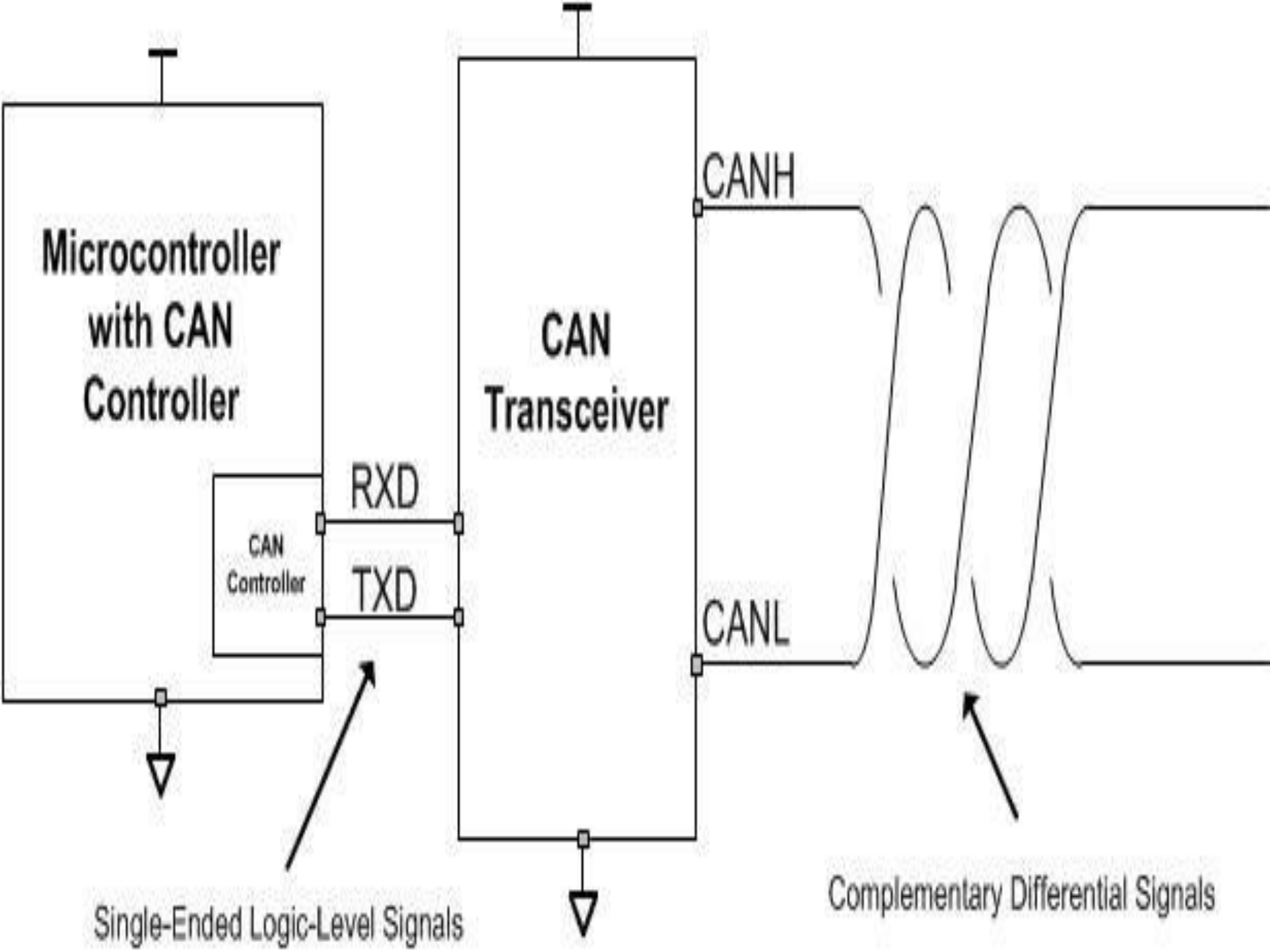
With CAN


Engine Control
CAN
Anti-Lock Brakes
CAN
Lighting
CAN
Air Con-dition
CAN
Power Locks
CAN
High Speed
CAN
Dash-board
CAN
Low Speed
CAN
Trans-mission Control
CAN
Active Suspen-sion
CAN
Power Seats
CAN
Power Win-dows
CAN
Airbag

Feature Cont'd

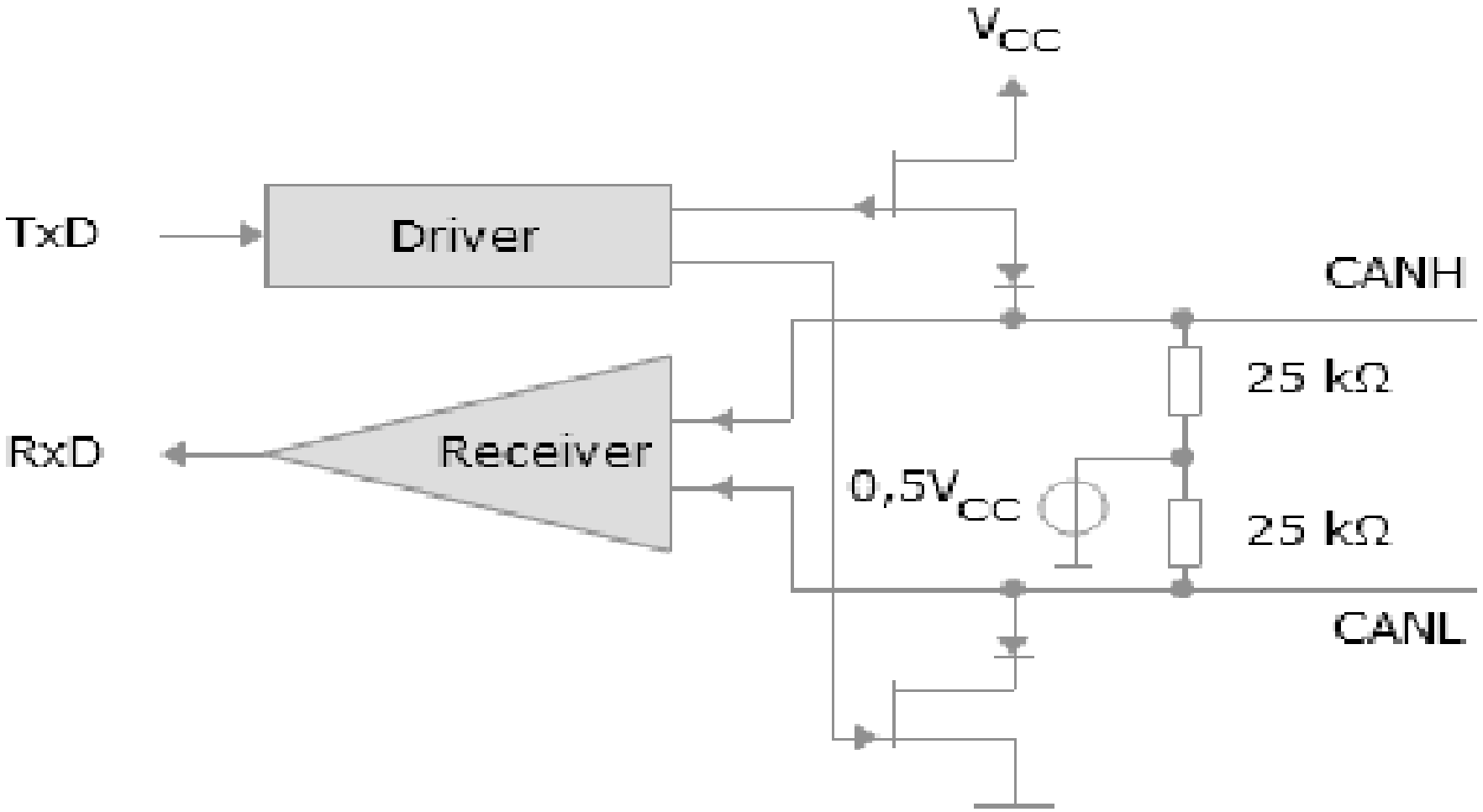
4. Supports 2 wired, twisted pair, differential bus, terminated at both ends with 120 ohm resistances.
5. The bus lines named as CAN_H & CAN_L.

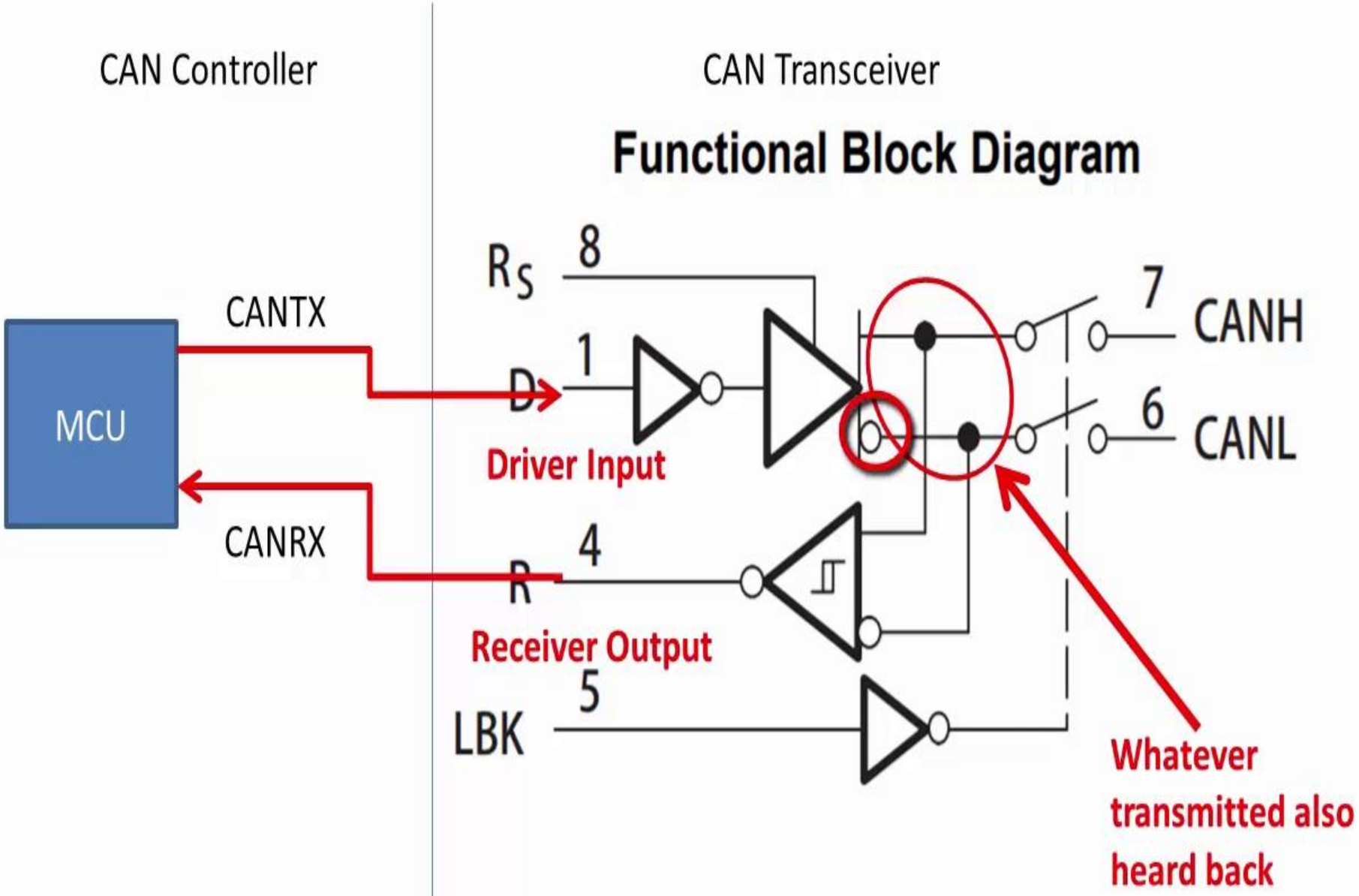



Microcontroller with CAN Controller
CAN Controller
RXD
TXD
CAN Transceiver
CANH
CANL
Single-Ended Logic-Level Signals
Complementary Differential Signals

CAN Communication

CAN Transceiver



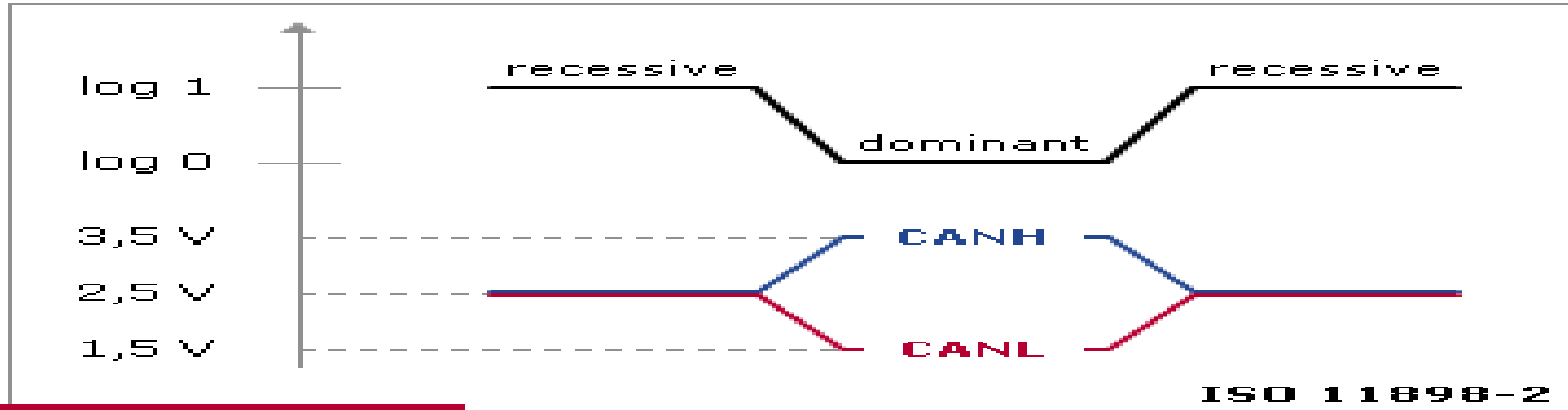

CAN Controller
CAN Transceiver
Functional Block Diagram
MCU
CANTX
CANRX
RS
8
D
1
Driver Input
R
4
Receiver Output
LBK
5
7
CANH
6
CANL
Whatever transmitted also heard back

Features Cont'd

6. The difference of positive voltages bt/w CAN_H & CAN_L is taken for logic '1'(recessive level) / '0'(dominant level).

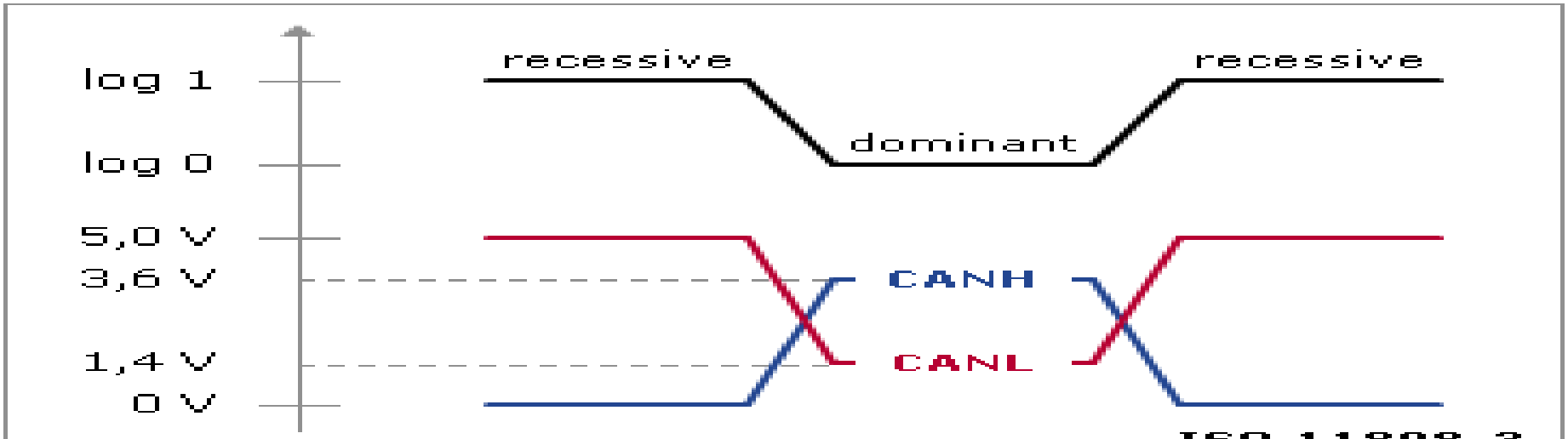
   E.g. Logic '1' <=0.5v & Logic '0' >=0.9v.
7. Supports asynchronous serial communication.
8. Data transfer order MSB first ONLY.

CAN Communication
High-Speed CAN Bus Levels



ISO 11898-2

CAN Communication
Low-Speed CAN Bus Levels



ISO 11898-3

FIGURE 4: ISO11898 NOMINAL BUS LEVELS

